Comparing OpenCV, Emugu CV and AForge.NET frameworks			
The difference	**OpenCV**	**Emugu CV**	**AForge.NET**
Licence	Free; BSD licence	Open source; GPL and commercial licence	Open source; LGPL v3
Documentation support	Strong support via books, websites, official documentation, samples, source code and forums	Less support	Less support
User interaction	Tough	Good	Very easy
Performance	Exceptional	Good	Poor
Grayscale image performance	Nice	Medium	Poor
Binarisation performance	Excellent	Good	Poor
Official website: *http://www.aforgenet.com* **Current version:** 2.25			

The framework comprises library sets and some sample applications that are listed below.

- *AForge.Imaging:* An image processing routines and filters library
- *AForge.Vision:* A computer vision library
- *AForge.Video:* A video processing library
- *AForge.Neuro:* A neural networks computation library
- *AForge.Genetic:* An evolutionary biology programming library
- *AForge.Fuzzy:* A fuzzy computations library
- *AForge.Robotics:* A robotics kits library
- *AForge.MachineLearning:* A machine learning library

Let's briefly discuss each of these libraries.

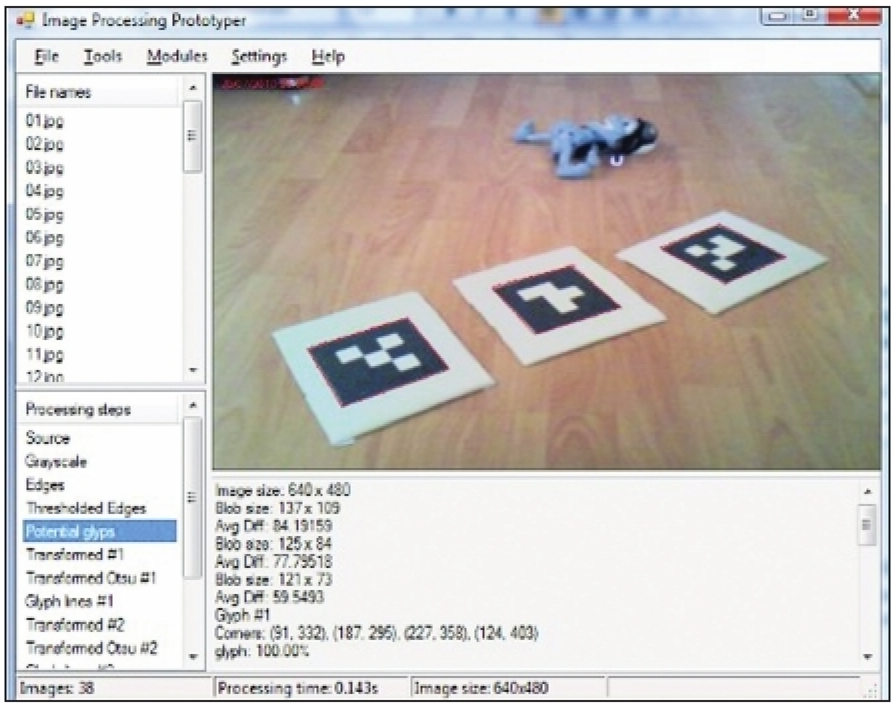

Figure 1: Image processing prototypers

AForge.Imaging

The image processing library contains a set of image processing filters and tools designed to address many different tasks of computer vision and image analysis/processing.

The library consists of the following filters:

- Colour filters (grayscale, sepia, invert, rotate channels, channel extraction, channel replacing, channel filtering, colour filtering, Euclidean colour filtering, RGB channels, linear correction, etc)
- HSL filters (linear correction, brightness, contrast, saturation, hue modifier and HSL filtering)
- YCbCr filters (linear correction, YCbCr filtering and channel extraction/replacement)
- Binarisation filters (threshold, threshold with carry, ordered dithering, Bayer dithering, Floyd-Steinberg, Burkes, Jarvis-Judice-Ninke, Sierra, Stevenson-Arce and Stucki dithering methods)
- Adaptive binarisation (simple image statistics)
- Mathematical morphology filters (erosion, dilatation, opening, closing, hit and miss, thinning and thickening)
- Convolution filters (mean, blur, sharpen, edges, Gaussian blur, sharpening based on Gaussian kernel, etc)
- Two source filters (merge, intersect, add, subtract, difference, move towards and morph)
- Edge detectors (homogeneity, difference, Sobel and canny)
- Gamma correction, median filter
- Conservative smoothing, jitter, oil painting, pixellate and simple skeletonisation
- Blob counter and connected components labelling filter
- Texture generators (clouds, marble, wood, labyrinth and textile), texturer, textured filter and texture merge filter
- Resize and rotation (nearest neighbour, bilinear and bicubic)
- Frequency filtering with FFT
- Image statistics
- Flat Field Illumination correction, simple skeletonisation, shrink, canvas crop/fill/move, mirroring, Bayer filter and mask/masked filter
- Fourier transformation (low-pass and high-pass filters)

AForge.Vision

This library consists of different motion detection and motion processing algorithms. The former are primarily targeted to detect motion in video frames by recording the amount of